



C fopen() function with Examples

Last Updated : 23 Jun, 2023

The **fopen() method** in C is a library function that is used to open a file to perform various operations which include reading, writing, etc. along with various modes. If the file exists then the fopen() function opens the particular file else a new file is created.

Syntax

The syntax of C fopen() is:

```
FILE *fopen(const char *file_name, const char *mode_of_operation);
```

Parameters

The method accepts two parameters of character pointer type:

- **file_name:** This is of C string type and accepts the name of the file that is needed to be opened.
- **mode_of_operation:** This is also of C string type and refers to the mode of the file access.

The below table lists valid mode_of_operation values in C with their meaning:

Opening Modes	Description
r	Searches file. <i>Opens the file for reading only.</i> If the file is opened successfully fopen() loads it into memory and sets up a pointer that

Opening Modes	Description
	points to the first character in it. If the file cannot be opened fopen() returns NULL.
w	Searches file. If the file exists already, its contents are overwritten. If the file doesn't exist, a new file is created. Returns NULL, if unable to open the file. <i>It creates a new file for writing only(no reading).</i>
a	Searches file. If the file is opened successfully fopen() loads it into memory and sets up a pointer that points to the last character in it. If the file doesn't exist, a new file is created. Returns NULL, if unable to open the file. <i>The file is opened only for appending(writing at the end of the file).</i>
r+	Searches file. <i>Opens the file for both reading and writing.</i> If opened successfully, fopen() loads it into memory and sets up a pointer that points to the first character in it. Returns NULL, if unable to open the file.
w+	Searches file. If the file exists, its contents are overwritten. If the file doesn't exist, a new file is created. Returns NULL, if unable to open the file. <i>The difference between w and w+ is that we can also read the file created using w+.</i>
a+	Searches file. If the file is opened successfully fopen() loads it into memory and sets up a pointer that points to the last character in it. If the file doesn't exist, a new file is created. Returns NULL, if unable to open the file. <i>The file is opened for reading and appending(writing at the end of the file).</i>
rb	Open the binary file in read mode . If the file does not exist, the open() function returns NULL.

[Read More](#)